

Deep Learning for Pneumonia Detection from Chest X-Ray Images

COMP263 Deep Learning Group Project – Section 001 – Group 3
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Abstract—This report presents a comparison of deep learning methods in the detection of pneumonia through the chest X-ray images. It conducted three experiments, custom CNN models, transfer learning using autoencoders, and ResNet50 using transfer learning vs from-scratch training. The results also suggest that Wide CNN attained the best balance of custom CNN and ResNet50 transfer achieved the best performance.

Keywords—Deep Learning, Chest X-Ray, Pneumonia Detection, Convolutional Neural Network, Autoencoder, Transfer Learning, ResNet50, Medical Image Classification.

I. INTRODUCTION (HEADING 1)

Pneumonia is a major cause of death in the world and early diagnosis with the help of the chest radiography is extremely important. Chest X-rays can only be interpreted manually, which is time-intensive and inter-observer variation occurs. Convolutional neural networks (CNNs) or deep learning have demonstrated the ability to be useful in automated image classification in the medical field.

The binary classification problem of pneumonia detection (Normal vs. Pneumonia) with the help of the chest X-ray images is discussed in this project. There are 5,216 training images and 624 test images in Kaggle Chest X-Ray Images (Pneumonia) dataset, approximately 75% of which are pneumonia. All the images were downresized to 224x224 and run through the ResNet50 preprocessing pipeline. The training set was again divided into 80/20 train-validation with random seed of 42 to be reproducible.

Three experiments were structured to systematically examine: (1) the role of CNN architecture depth and width in determining classification performance, (2) the value of unsupervised pre-training with autoencoders to extract features and (3) the value of transfer learning with an ImageNet using a state-of-the-art ResNet50 architecture.

II. METHODOLOGY

Experiment 1 – Custom CNN Architectures: Three CNN versions were built and trained by hand. The Baseline model is based on three convolutional blocks [32, 64, 128] and a dense layer with 256 units. Deep model adds an additional block (filters [32, 64, 128, 256]) to the network to make it deeper. The Wide model is used to increase capacity, by using larger filters [64, 128, 256] and a dense layer with 512 units. Each model has 3x3 convolutions, BatchNormalization, ReLU, MaxPooling, GlobalAveragePooling, L2 regularization (0.001), and 50% dropouts. Random horizontal flipping, rotation (± 0.1) and zoom (± 0.1) are all examples of data augmentation. Adam optimizer (lr=0.001), categorical cross-entropy loss, EarlyStopping (patience=4) and ReduceLROnPlateau (factor=0.5, patience=2) were trained over a maximum of 15 epochs.

Experiment 2 – Autoencoder and Transfer Learning: A convolutional autoencoder was unsupervisedly trained to acquire compact feature representations of X-ray images. The encoder consisted of VGG-style Conv2D, BatchNorm, ReLU, MaxPooling and dropout layers, and a bottleneck with 256 filters. Images were rebuilt by the decoder based on the UpSampling2D and linear output, which was trained with the MSE loss over 20 epochs. Phase 2 Phase 2 involved transfer of the pre-trained encoder weights to a classifier consisting of Conv2D(512), GlobalAveragePooling, dense layers (1024, 512), BatchNorm and dropout. Adam (lr=0.0005) was used to fine-tune the entire network with 15 epochs.

Experiment 3 – State-of-the-Art ResNet50: There were two variants of ResNet50. The transfer learning model takes ImageNet pre-trained weights and applies a custom head (Dense 512, 50% dropout, softmax output) to the weights and trains using Adam (lr=0.001). The from-scratch model is configured with random initialized weights of ResNet50, heavier augmentation with stronger data augmentation, a deeper classifier head with BatchNormalization, Dense (1024), Dense (512), and dropout layers, and a lower learning rate for stable training.

III. RESULT

All models were tested using the held-out (test set) of 624 images which are 234 normal and 390 pneumonia. Table presents results of custom CNN architecture of Experiment 1.

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TABLE I. Experiment 1 – Custom CNN Results (Test Set)

Model	Accuracy	Precision	Recall	F1 Score
Baseline	0.5545	0.9746	0.2949	0.4528
Deep	0.7837	0.7918	0.8872	0.8368
Wide	0.7965	0.7982	0.9026	0.8472

The Baseline CNN is very precise (0.975) but very low in recall (0.295), which implies that it predicts the majority of samples to be Normal. Deep and Wide models enhance considerably the recall and the balance in general. The Wide CNN has the highest F1 score (0.847) of all custom models, showing that more filter width and dense layers capacity are more useful to this binary classification task than added depth.

Table II presents the results of Experiment 2 in which the feature extraction was performed with the help of the autoencoder-based unsupervised pre-training.

TABLE II. Experiment 2 – Autoencoder + Transfer Learning (Test Set)

Model	Accuracy	Precision	Recall	F1 Score
Autoencoder Transfer	0.7580	0.7234	0.9923	0.8368

The autoencoder model recalls almost all the cases of pneumonia (0.992), i.e. it can identify almost all cases of pneumonia but with a lower precision (0.723) as it has a high false positive rate (148 of 234 Normal images misclassified). This trade-off may be applied in clinical screening whereby it is costly to fail to create an alarm on pneumonia than a false alarm.

TABLE III. Experiment 3 – ResNet50 SOTA (Test Set)

Model	Accuracy	Precision	Recall	F1 Score
ResNet50 From-Scratch	0.6250	0.6250	1.0000	0.7692
ResNet50 Transfer	0.8157	0.7944	0.9513	0.8658

ImageNet transfer learning on ResNet50 showed the overall best performance, with the highest F1 score (0.866) and accuracy (0.816) of all the experiments. This demonstrates that trained features are easily transferred to the medical imaging field. Conversely, the from-scratch model suggested the majority of samples as pneumonia (recall = 1.0, precision = 0.625), indicating how challenging it is to train

large (23 million parameters) networks on a small dataset without pre-training. Transfer learning increased the accuracy by 19.07% over training on fresh. In general, both the transfer learning methods performed better than random-initiation, with ResNet50 transfer demonstrating the most appropriate performance balance, and the autoencoder model being the most sensitive, using in clinical screening.

IV. CONTRIBUTION

Contribution The work of each group member is as follows:

- 1) Drashtam Kinnar Banker: Designed, trained, and evaluated the Baseline CNN model in Experiment 1.
- 2) Zeel Rameshbhai Vekariya: Designed, trained, and evaluated the Deep CNN model in Experiment 1.
- 3) Juan David Barrero Guerrero: Designed, trained, and evaluated the Wide CNN model in Experiment 1.
- 4) Jaturaput Jongsubcharoen: Designed and implemented the Autoencoder model and encoder-based transfer learning pipeline in Experiment 2.
- 5) Alim Rashyani: Implemented and evaluated the ResNet50 From-Scratch model in Experiment 3.
- 6) Raj Patel: Implemented and evaluated the ResNet50 Transfer Learning model in Experiment 3.

All group members contributed to README documentation, discussion, and final presentation preparation.

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